Show the details of your work for full credit.
Write your answers clearly and concisely.

1. Find the domain of $f(x)$.

(a) $f(x)=\sqrt{\dfrac{x-1}{x^2+x-6}}$

(b) $f(x)=\dfrac{1}{x^2+4x-21}$

(c) $f(x)=\dfrac{\sqrt{x^3-x^2-9x+9}}{3x+2-\sqrt{x^2-4}}$

2. (a) State the definition of the derivative of a differentiable function $f(x)$ at $x=a$.

(b) Find the limit $\lim\limits_{h\to 0}\dfrac{(3+h)^2-9}{h}$.

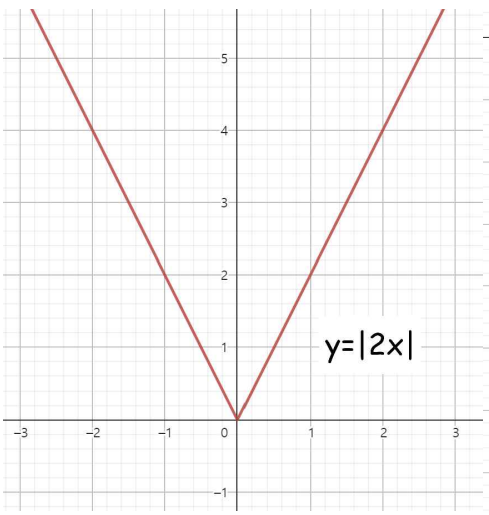
(c) Find a function $f(x)$ and the value of a such that the derivative of f at a is equal to the result obtained in part (b).

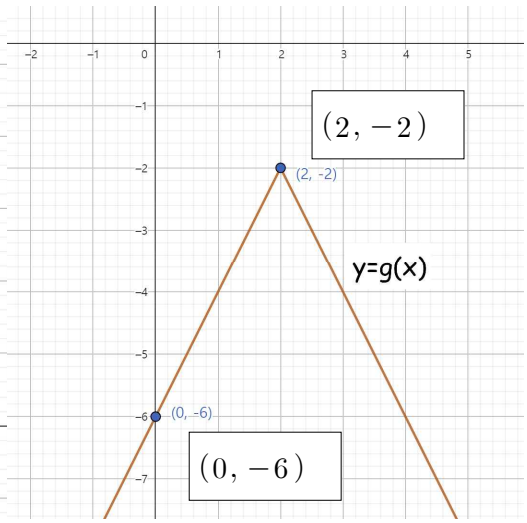
3. Find the limit of the following, if it exists.

(a) $\lim\limits_{x\to 2}\sqrt{x^2-4}$

(b) $\lim\limits_{x\to 4}\dfrac{7-\sqrt{12x+1}}{x-4}$

4. Consider the graph of the function $f(x)=|2x|$.
Determine the expression for $g(x)$ in the second figure using the absolute value function.





5. Let $f(x)=\dfrac{x}{x^2+1}$.

(a) Find the derivative of f.

(b) Find the vertical and horizontal asymptotes, if they exist.

(c) Find the intervals on which f is increasing and intervals on which f is decreasing.

(d) Find the second derivative of f.

(e) Find where the graph of f is concave upward and where it is concave downward.

(f) Find the points (x,y) where $f(x)$ has the local extrema or an inflection point. Specify each point with its type.

(eg. (x_0, y_0) : local maximum, (x_1, y_1) : local minimum)

6. If $f''(x)=24x^2-3x-\dfrac{3}{2}$, $f(0)=-1$ and $f'(0)=2$, find f.